Jonathan Tobias

Robotics/Electronics Engineer
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Work Permit Status: Currently in NL on a Highly Skilled Migrant Visa (valid until 2030)

Summary

Hands-on Robotics & Electronics Engineer with over 5 years of experience bridging hardware, firmware, and software across agricultural and automotive environments. Collaborative team player and lifelong learner with a proven track record of developing hardware and software solutions across independent, team-driven, and personal projects. I am seeking on-site or hybrid engineering roles in the Netherlands.

Technical Highlights

- **Robotics & Software:** ROS1 / ROS2, Python, C++, Behavior Trees, Docker, Git, Linux, Computer Vision, SQL
- **Design Tools:** Altium Designer, KiCAD, EasyEDA, SolidWorks, and Onshape.
- **Firmware & Embedded Devices:** Embedded C, STM32, ESP32, ATxmega, Raspberry Pi, RDK X5
- **Protocols & Systems:** CAN bus, I2C, SPI, RS232, LoRa, RS485, TCP, UDP and MQTT.

Work

Lensor Ltd. | Netherlands
Integration Engineer | March 2025 – Present

- Designed and prototyped custom Raspberry Pi PCBs to interface with external sensors and low-voltage systems.
- Redesigned a custom sensor layout for vehicle tracking and circuitry to extend I2C communication far beyond standard operating distance specifications.
- Designed and built an automated self-test jig for hardware validation, significantly accelerating production testing for custom PCBs.
- Developed Python-based R&D software features and solutions, integrating camera systems, MQTT messaging, Docker containers, and Sentry logging.
- Networking for multi-camera setups (+10 cameras)

Axis7 Ltd. | New Zealand
Robotics Engineer | February 2022 – June 2024

- Developed software in Python around the ROS framework for automated agricultural machinery, including an in-house commercial broccoli harvester and custom 4-wheeled autonomous arch vehicles.
- Integrated complex sensor suites (RTK-GPS, LiDAR, IMU, Stereo/ToF cameras, PLCs, RF receivers) over CAN bus using Behavior Tree logic and an extended Kalman filter package to combine the mentioned sensors.
- Implemented and trained our own object detection using open-source computer-vision models to improve machine perception and field navigation (PyTorch, OpenCV, and YOLO [Keypoint, Segmentation] models).

Bentech Ltd. | New Zealand
Electronic Engineer | November 2017 – November 2020

- Designed, programmed, and self-managed custom PCB projects from concept to deployment, utilizing Altium and embedded C (ATxmega & ESP32).
- Developed ultra-low-power remote sampling units with Wi-Fi web interfaces, wireless temperature monitoring systems, and infrastructure automation circuits.
- Electronics communication using LoRa, RS232, RS485, I2C, SPI, and CAN bus protocols.
- Modeled mechanical enclosures using SolidWorks to house custom PCB assemblies.

Education and Research

Master of Engineering (First-Class Honors)
The University of Waikato, New Zealand | Nov 2020 – Mar 2022

- **Thesis:** Developed a transferable semi-autonomous vehicle architecture for navigation within corridor crops (vineyards and orchards).
- **Skills Applied:** ROS1, Python, C++, CAN Bus, Hardware-Software System Design.
- **Video:** A small video of testing and experimenting on the project.
- **Publication:** Transferable semi-autonomous architecture for vehicles operating in corridor crop environments

Bachelor of Electronic Engineering (First-Class Honors)
The University of Waikato, New Zealand | Feb 2014 – Apr 2018

Language, Hobbies and Sporting Achievements

- **Languages:** English (Native / Fluent), Dutch (A2–B1 Level / Actively Learning)
- **Sports & Achievements:**
    - Represented New Zealand at the Tertiary Futsal World Cup (Kazakhstan, 2018).
    - 2x Winner of the NZ Futsal Championship with the Waikato Men's Team.
    - I enjoy playing football, marathon running, hiking, and bouldering.